

Final practice set: function decoders and filters

PN2, Lectures 4–5

Questions. Solutions, with plots, are in a separate file. Leave that file closed.

How to use this

Every question uses a setup you have not seen in the earlier sheets. The skills are the same: build A and the target, solve for a decoder, push it into weights, and reason about what a synaptic or optimal filter does to a signal. Sketch whenever a question says “sketch”, with labelled axes. No calculator is needed beyond the numbers below.

Useful numbers

$e^{-1} \approx 0.368$, $e^{-2} \approx 0.135$, $e^{-3} \approx 0.050$, $e^{-0.25} \approx 0.779$, $\pi \approx 3.14$, $\ln 10 \approx 2.30$,
 $\sqrt{0.9} \approx 0.949$, $-\ln(0.051) \approx 2.97$, $\sqrt{1.394} \approx 1.181$, $\sqrt{40.4} \approx 6.36$.

Formulas you may quote

Function decoder: $(D^f)^\top = (AA^\top + N\sigma^2 I)^{-1}AF^\top$, column k of F is $f(x_k)$.

Weights: $W^f = ED^f$, row i of E is $\alpha_i e_i^\top$, $J = W^f a + J^{\text{bias}}$.

Normalized PSC: $h_n(t) = t^n e^{-t/\tau} / (n! \tau^{n+1})$ for $t \geq 0$, $|H_n(\omega)| = (1 + \omega^2 \tau^2)^{-(n+1)/2}$, mean delay $(n+1)\tau$.

1 Decoders from kinked tuning curves

[function decoders]

Three rate neurons represent a scalar $x \in [-1, 1]$:

$$a_1 = \max(x, 0), \quad a_2 = \max(-x, 0), \quad a_3 = \max(x + 1, 0).$$

Train on the three inputs $x = (-1, 0, 1)$.

- Form A (rows are neurons, columns are samples). Sketch the three tuning curves on $[-1, 1]$. Which neuron has its kink inside the interval, and which neuron is a straight line on the whole interval?
- Find exact decoders, by matching all three samples, for $f(x) = x$, $f(x) = |x|$, and $f(x) = 1$. Then explain in one sentence why the unregularized least-squares formula would give the same answers here.
- Without solving anything new, write the decoder for $f(x) = 3x - 2$. Verify it at $x = 1/2$, which is not a training input.
- Now use the target $f(x) = x^2$ on the same three samples. Which decoder do you get? Evaluate the decoded value at $x = 1/2$ and compare with $(1/2)^2$. What went wrong?
- Show that for any decoder (p, q, r) the decoded value on $[-1, 1]$ has the form $c_0 + c_1 x + c_2 |x|$. Fit x^2 over the whole interval instead of three samples: by symmetry $c_1 = 0$, so fit $c_0 + c_2 x$ to x^2 on $[0, 1]$ in least squares. Convert the result into a decoder (p, q, r) . Where is its error largest?

2 What the noise penalty buys

[regularized decoder]

A single neuron fires at $a = 1$ when $x = 1$ and at $a = 2$ when $x = 2$. Decode x with one coefficient d , so $\hat{x} = da$. There are $N = 2$ samples.

- Find the unregularized decoder. What are the decoded values at the two samples?
- Find the regularized decoder for $\sigma^2 = 2.5$. What are the decoded values now? In which direction did the decoder move?
- Suppose each recorded rate carries independent noise of variance $\sigma^2 = 2.5$, so the decoded value carries noise of variance $d^2\sigma^2$. Write the expected squared error $E(d)$ as the clean error averaged over the two samples plus this noise term. Evaluate $E(1)$ and $E(1/2)$. Which is smaller?
- Minimize $E(d)$ and show the minimizer is the decoder you found in (b). Sketch the clean error, the noise error, and the total against d .

3 A two-dimensional function, then weights

[vector decoders]

Reuse the population from Question 1. The connection should send the vector $y = f(x) = (x, |x|)^T$ to a population B of three neurons:

B neuron	gain α	unit encoder e	J^{bias}
1	2	(1, 0)	1/2
2	1	(0, 1)	1
3	3	(-1, 0)	0

- Write D^f using your answers to Question 1(b). State its shape.
- Write E and state its shape. Then compute $W^f = ED^f$ and state its shape.
- One column of W^f is zero. Which one, and why does that pre-neuron get no weight for this function?
- At $x = -1$, compute B's three currents twice: from $W^f a + J^{\text{bias}}$, and from $\alpha_i \langle e_i, y \rangle + J_i^{\text{bias}}$ with the true y . They should agree.
- Dale's principle says each pre-neuron should be either excitatory or inhibitory onto all of its targets. Which pre-neurons violate it in W^f ?

4 An exponential synapse on a regular spike train

[synaptic filter]

Use $h_0(t) = e^{-t/\tau}/\tau$ for $t \geq 0$, with $\tau = 5$ ms. The filtered activity is $s(t) = \sum_k h_0(t - t_k)$.

- A neuron spikes at $t = 0, 5, 10$ ms and then stops. Compute $s(10^-)$, $s(10^+)$, and $s(15^-)$ with units. Sketch $s(t)$ from 0 to 30 ms.
- Now the neuron spikes every $T = 5$ ms forever. In steady state, every spike lands on the same trough and jumps to the same peak P . Use $P = Pe^{-T/\tau} + 1/\tau$ to find P and the trough.

- (c) Show that the average of $s(t)$ over one steady-state period is exactly $1/T$. Which property of h_0 makes this true?
- (d) Repeat (b) with $\tau = 20$ ms. Compare the peak-to-trough ripple and the mean with the $\tau = 5$ ms case. With a decoder coefficient $d = 0.004$, what are the decoded mean and ripple for each τ ? What does the larger τ cost?

5 Two synapses with the same delay

[filter shape]

Filter A is h_0 with $\tau = 10$ ms. Filter B is h_1 with $\tau = 5$ ms.

- (a) Show that both filters have the same mean delay.
- (b) Compute $|H|$ for both filters at $f = 10$ Hz (a signal frequency) and at $f = 100$ Hz (where spike noise lives). Use $\omega = 2\pi f$.
- (c) Which filter is better on both counts? Sketch both kernels on common axes, and state one way filter B responds worse to a sudden change.
- (d) The input is bandlimited to 10 Hz. Why does the spike train still have power at 100 Hz?

6 The optimal filter when the response is amplified [optimal filter]

The signed response of an opponent pair is $R = gX + \eta$, where $g > 0$ is a gain, $S_X = \langle |X|^2 \rangle$, $S_\eta = \langle |\eta|^2 \rangle$, and $\langle X\eta^* \rangle = 0$.

- (a) Expand $H = \langle XR^* \rangle / \langle |R|^2 \rangle$ to get H in terms of g , S_X , S_η .
- (b) What does H become when $S_\eta \rightarrow 0$? When $S_X = 0$? Interpret both.
- (c) Let $S_X = 1$ for $|f| \leq 5$ Hz and 0 otherwise, with $S_\eta = 4$ at all frequencies. Compute the in-band H for $g = 1, 2, 4$ and the out-of-band H .
- (d) The input reaches the output through gH . Compute gH for the three gains. Why can $g = 1$ and $g = 4$ have the same H but very different output quality? Sketch gH against f .

7 Filter first or square first?

[filters in a chain]

Population C should represent x^2 . The input $x(t)$ steps from 0 to 1 at $t = 0$. Assume every decoder is exact, neurons respond instantly, and every synapse is h_0 with the same τ , so a step passed through one synapse becomes $1 - e^{-t/\tau}$.

Route 1 connects A to C with a decoder for x^2 . Route 2 connects A to B with the identity, then B to C with a decoder for x^2 .

- (a) Route 1: write the signal entering C and evaluate it at $t = \tau$.
- (b) Route 2: write the value B represents, and the value B's x^2 decoder produces, before the second synapse. Evaluate the second at $t = \tau$.
- (c) Both routes decode x^2 , so why do (a) and (b) differ? When would they agree?

- (d) After the second synapse, Route 2 delivers $1 - 2(t/\tau)e^{-t/\tau} - e^{-2t/\tau}$ to C. Evaluate it at $t = \tau$ and rank the three signals.
- (e) Find the time for Route 1 to reach 0.9, and the time for B's decoded x^2 in Route 2 to reach 0.9. Sketch all three signals against t/τ .